JIGNESH MOTWANI

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EDUCATION

Singapore University of Technology and Design (SUTD) **Sep 2021 – May 2025**

- Bachelor of Engineering in **Computer Science and Design**, Honours with Merit
- **Minor in Artificial Intelligence**, Specialization Tracks: Data Analytics, Financial Technology
- **Awards & Leadership:** Undergraduate Merit Scholarship; Best Student Project Award, SUTD Machine Learning (third-order HMM + Viterbi, top of 50+ teams); Teaching Assistant, SUTD Algorithms (300+ students)

SKILLS

Languages: Python, TypeScript, JavaScript, Java, Golang, C++, SQL

AI & GenAI: LLMs, RAG, Agentic AI & Multi-Agent Systems, Fine-tuning (LoRA/PEFT), LLM Evaluation, Computer Vision, PyTorch, LangGraph, LangChain, MCP

AI Platforms & LLMOps: AWS Bedrock (AgentCore, Knowledge Bases), Gemini Enterprise Agent Platform (Vertex AI), Azure AI Foundry, LiteLLM, Langfuse, OpenTelemetry, OPA, pgvector

Technologies & Tools: React, Node.js, FastAPI, Docker, Kubernetes, Terraform, PostgreSQL, AWS, Azure, GCP, GitLab CI/CD

WORK EXPERIENCE

AI Engineer (Data & AI Team), Keppel Technology Solutions **Jan 2026 – Present**

- Lead a 6-person engineering team on K2, an enterprise AI agent platform used by **300+ employees**, as its **product owner and architect**; review and merge code and deploy via GitLab CI/CD to UAT and production on Kubernetes.
- Build K2's platform-agnostic **agent lifecycle and governance**: generate, sandbox-test, certify and release Python (Strands) agents, today on AWS Bedrock AgentCore, with OPA policy gates, LiteLLM budgets and Langfuse tracing.
- Co-develop an **AIOps platform** for incident, compliance and change management on AWS and Azure, **sold to multiple clients**: correlates alerts, suggests root causes via RAG over runbooks and applies human-approved fixes with rollback.
- Own its service graph, compliance auto-fixes, CVE handling and ServiceNow integration, against MVP targets of **50%+ less alert noise**, 80%+ top-3 root-cause accuracy and **under $1** of AI cost per incident.
- Deliver GenAI solutions directly with enterprise clients (**10+ to date**) on AWS Bedrock, Azure AI Foundry and Gemini Enterprise Agent Platform: RAG assistants, agentic workflows, voice AI and computer vision, with **LLM-as-judge evals**.

Full Stack Algorithms Engineer Intern, FireJet **May 2024 – Sep 2024**

- Developed and optimized a one-click **Figma-to-code converter** (React, Vue and Tailwind output) in TypeScript and Node.js, helping grow platform users **from 30k to 60k**.
- Integrated design-to-code engine improvements into the new Website-to-Figma product, focusing on output accuracy; it reached **5k users in its first month**.
- Used Supabase, RESTful APIs and custom algorithms to improve design-to-code conversion across FireJet's products.

Software Engineering Intern, pQCee Pte Ltd. **Aug 2023 – Mar 2024**

- As **sole developer**, built **QuantumNFT** end to end: a Web3 platform on Bubble where developers showcase quantum programming skills through NFT-based coding challenges ("Kaggle for Quantum Programming").
- Wrote custom **JavaScript plugins** for Bubble that convert quantum circuits into uniquely minted NFTs, with MetaMask wallet integration.
- Took the platform from **v0.1 to v0.6 and onto testnet** through agile iterations with test-driven development.

PROJECTS

P2P File Sharing System | **Github**

- Built a **distributed peer-to-peer file sharing system** from scratch in Golang with no external libraries, on the Chord DHT protocol with consistent hashing for file storage and retrieval.
- Handled fault tolerance, including **Byzantine faults**, dynamic node joins and departures, and simultaneous failures.

Live Sign-Language Translation using Computer Vision | GitHub

- **Led a team of 4** to build real-time sign-language-to-text translation with computer vision and Google MediaPipe, reaching **99.5% accuracy**.
- Developed the hybrid Point-CNN and PointNet deep learning model behind the translation.

Kiora Care

- **Architected and built KIORA**, a healthcare platform for patients, doctors and clinics to manage dialysis (CAPD) sessions, vitals, medications, inventory and appointments.
- Built with FastAPI, Node.js, React and PostgreSQL in Docker, deployed on AWS (ECS/Lambda, RDS, S3) with role-based access control and notifications.

CERTIFICATIONS

- **AWS Certified Generative AI Developer – Professional | Credential**
- **Microsoft Certified: Azure AI Apps and Agents Developer Associate | Credential**